

Fragrance Demand Analysis and Recommendation for Aura Kesidang Using Machine Learning

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Abstract—This proposed study suggests a data-driven model to determine high potential fragrance segments for Aura Kesidang business expansion. This analysis goes beyond traditional measures like sales volume or unit price, whereby the practice measures market potential using integrated indicators of demand such as gender preference, per unit category and rating. The solution focuses on scalability and strategic fit, allowing the SME to invest in products with sustainable growth trajectories, instead of short-term operational optimization.

Keywords—fragrance, high, demand

I. INTRODUCTION

A. Background of Study

Aura Kesidang is a small-to-medium enterprise (SME) manufacturing natural essential oils such as Kesidang, Peppermint, Cengkih, and Lemongrass, supplying the market with wellness and aromatherapy products. Products are distributed via independent agents across regions, with a daily record of sales, price, and inventory data. The initial breakdown of the dataset shows total sales of 96 units across all variants with a revenue of RM1,536: Peppermint (31 units) and Kesidang (26 units) are the highest selling variants.

B. Problem Statement

Although the unit sales are high, Aura Kesidang lacks insight for products that give the greatest returns per unit of production effort. For instance, Lemongrass sells (25 units), yet its stock hit zero by 10/13/25, indicating that it might have supply constraints or high demand. Meanwhile, Cengkih (14 units sold) has a relatively high stock (49 units), indicating weaker turnover. Decisions based on sales alone can be a misallocation of resources, particularly in cases where packaging time, unit cost, market demands, and regional agent performance are not considered.

C. Objectives of the Study

This paper intends to develop a data-driven decision analysis for Aura Kesidang to determine a new flavor for their oil products in order to expand their business. Specifically, to examine SME's dataset with public dataset identifying which fragrance lines are yielding the highest market potential.

D. Proposed Solution

The proposed solution suggests a decision-support analysis of the SME's current dataset, augmented with public dataset, to compute key measures: unit demand level for each flavor based on total and average ratings, fragrance category, and gender preference; to recommend a new

flavor. By integrating these data with visuals, Aura Kesidang can shift from predictive to reactive optimized planning, knowing which fragrance category and flavor have the highest market potential.

II. LITERATURE REVIEW

A. Comparison with Existing Solutions

While other alternatives like price-volume analysis, profit estimation or production-efficiency evaluation provide partial insights, they are either too narrow (e.g., prioritizing Peppermint to the highest unit sales regardless of its low average RM 49.5/unit), lacking data (e.g., profit margins need unrecorded costs data), or not aligning with strategic growth (e.g., optimization of packaging time on Cengkih which has declining demand, inventory rose from 41 to 49 despite only 14 units are sold). In contrast, our proposed solution evaluates the market potential based on integrated signals (unit demand, ratings, gender preference and scent category) from both SME and public dataset to determine scalable opportunities allowing resource-efficient, demand driven scaling.

III. ANALYSIS

A. Dataset Preprocessing

1) Aura Kesidang Sales (SME Dataset) - Missing Values Identification

The SME dataset is actual business sales data of four types of fragrances, i.e. Kesidang, Peppermint, Cengkih and Lemongrass. It incorporates sold quantity, stock level, unit price, total price and date of transaction. This data is vital in the explanation of SME sales performance and inventory behavior.

The missing values were detected with the help of the following function: `df.isna().sum`, it showed that there were some missing values in the stock-related columns.

```
# Check Missing Values
print("Missing values before cleaning:\n", sales_df.isnull().sum())
```

Missing values before cleaning:

Date	0
Kesidang_Sold	0
Peppermint_Sold	0
Cengkih_Sold	0
Lemongrass_Sold	0
UnitPrice	0
TotalPrice	0
Kesidang_Stock	8
Peppermint_Stock	8
Cengkih_Stock	8
Lemongrass_Stock	8
dtype:	int64

Figure 1: SME missing values BEFORE cleaning Missing Values

Median imputation was applied using `df.fillna(df.median())` to replace missing numeric values. Median was chosen

because it is robust to outliers and preserves data distribution.

```
# Handle Missing Data (use median instead of drop)
sales_df = sales_df.fillna(sales_df.median(numeric_only=True))

# Confirm missing handled
print("\nMissing values after filling:\n", sales_df.isnull().sum())
```

```
Missing values after filling:
Date          0
Kesidang_Sold 0
Peppermint_Sold 0
Cengkih_Sold 0
Lemongrass_Sold 0
UnitPrice     0
TotalPrice    0
Kesidang_Stock 0
Peppermint_Stock 0
Cengkih_Stock 0
Lemongrass_Stock 0
dtype: int64
```

Figure 2: Missing values AFTER cleaning

2) SME Dataset Reshaping

The SME data was initially in the wide format. It was transformed into long form with the help of `pd.melt` to ensure that the fragrance types are displayed as values in a single column. This reorganization enables the compatibility with the popular dataset.

SME dataset first 10 rows after reshaping:

```
#Sort by date
sales_df = sales_df.sort_values('Date').reset_index(drop=True)
print(sales_df.head(10))
```

	Date	Year	Flavor	Category	Total_Ratings	Avg_Rating	Gender
0	2025-09-11	2025	Kesidang	Floral	1	4.9	women
1	2025-09-11	2025	Lemongrass	Fresh	2	4.0	unisex
2	2025-09-11	2025	Peppermint	Fresh	3	4.9	women
3	2025-09-11	2025	Cengkih	Spicy	1	4.8	female
4	2025-09-12	2025	Cengkih	Spicy	1	4.4	men
5	2025-09-12	2025	Kesidang	Floral	1	4.9	men
6	2025-09-12	2025	Lemongrass	Fresh	2	3.9	women
7	2025-09-12	2025	Peppermint	Fresh	3	3.9	unisex
8	2025-09-13	2025	Kesidang	Floral	2	4.7	unisex
9	2025-09-13	2025	Peppermint	Fresh	5	4.5	men

Figure 3: SME dataset first 10 rows after reshaping

3) Public Fragrance Trend Dataset (Fragrantica)

The publicly available data includes more than 24,000 records of fragrance trends on the internet based on Fragrantica. It comprises name of perfume, brand name, rating value, rating number, gender, perfumer, year, smell notes (top, middle, base). This information gives fragrance preferences on a market level.

4) Initial Dataset Inspection

`df.head()`, `df.info` and `df.describe` were used to test the data structure/quality of the dataset. These functions present a type of data, missing values and summary.

```
Display basic information of the columns:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24063 entries, 0 to 24062
Data columns (total 18 columns):
#   Column          Non-Null Count  Dtype
---  -
0   url              24063 non-null  object
1   Perfume          24063 non-null  object
2   Brand            24063 non-null  object
3   Country          24063 non-null  object
4   Gender           24063 non-null  object
5   Rating Value     24063 non-null  object
6   Rating Count     24063 non-null  int64
7   Year             22026 non-null  float64
8   Top              24063 non-null  object
9   Middle           24063 non-null  object
10  Base             24063 non-null  object
11  Perfumer1        24063 non-null  object
12  Perfumer2        1336 non-null  object
13  mainaccord1      24063 non-null  object
14  mainaccord2      24050 non-null  object
15  mainaccord3      23949 non-null  object
16  mainaccord4      23675 non-null  object
17  mainaccord5      23082 non-null  object
dtypes: float64(1), int64(1), object(16)
memory usage: 3.3+ MB
None
```

Figure 4: Dataset Inspection (Public Dataset)

5) Public Dataset Cleaning

Missing values before cleaning:

```
url          0
Perfume      0
Brand        0
Country      0
Gender       0
Rating Value 0
Rating Count 0
Year         5981
Top          0
Middle       0
Base         0
Perfumer1    0
Perfumer2    70434
mainaccord1  0
mainaccord2  38
mainaccord3  305
mainaccord4  1002
mainaccord5  2474
Flavor       0
dtype: int64
Before remove missing values: [74854 rows x 5 col]
After remove missing values: [68873 rows x 5 col]
Gender       0
Rating Value 0
Rating Count 0
Year         0
Flavor       0
dtype: int64
```

Figure 5: Public dataset missing values BEFORE cleaning

`df.drop` was used to eliminate irrelevant columns and the rows that had missing critical values were eliminated. This guarantees that only comprehensive and pertinent records are left.

B. Data Preprocessing & Feature Engineering

Cleaning: Removes the presence of those values in the data that lack real values.

- SME data: median imputation.
- Public data: deletion of partial records and normalization of text.

Feature Extraction:

- Date columns extracted year, month and day.
- Segregate the smell into separate flavors.
- Re regular expression cleaned text.

Category Mapping:

- The values of scent were categorized. The SME categories are Floral, Fresh, and Spicy, and the public is Floral, Fruity, Woody, Amber, Fresh, Spicy, and Gourmand.

Filtering:

- The flavors were eliminated that are not common and only the years 2018–2024 were kept to keep the records relevant and consistent.

Enhancement and Data fusion:

- The data sets of the SME and the public were concatenated. Other qualities like gender, average rating (3.5-5.0) and dates were also created synthetically so as to balance the datasets.

C. Technologies in Data Processing.

Pandas: The use of pandas is in data loading, cleaning, reshaping, filtering, and merging.

NumPy: Numbers and augmenting random data.

Regex (re): Text cleaning and scent extractions.

Scikit-Learn: Scikit-Learn is applied to preprocessing, encoding, standardization and machine learning models.

Matplotlib & Seaborn: Applied to visualize the distributions and trends of the scents.

IV. RESULT AND DISCUSSION

1) Clustering for Demand Label Identification

Clustering was used to find hidden demand patterns in the fragrance dataset as a preprocessing step. Since the original dataset did not have explicitly-labeled demand levels, the clusters obtained were added as a new variable, called Demand Level, and these served as dependent variables in two classification models: 1) Decision Tree, and 2) Random Forest, with the purpose of allowing their predictive performances to be compared.

For encoding, Category and Gender are two non-numeric features; these features need to be encoded in a format that will compatible with the clustering algorithm. Other non-numerical attributes like Flavor were not encoded since they are more descriptive features and not the main factors for cluster formation. In the clustering process, the encoded variables were accorded variable names like CategoryCode and GenderCode to ensure that it would be easy to understand the interpretation of the variables.

Feature scaling has been done to ensure that all the variables will contribute proportionately in clustering algorithms, and variables with a larger numerical range, like Total Ratings, will not dominate the distance computation. The average rating is used to record the different ratings provided by the customers, and the scale of the GenderCode and CategoryCode transforms categorical data to numerical data with a similar scale.

Subsequently, the K- Means algorithm had to be used in order to group fragrances based on their similarities in the scaled feature space. The Elbow Method was used to decide the optimum number of clusters by analyzing the inertia of different K values, 2 to 8. The stability value of the curve was around K = 4 indicating 4 cluster groups.

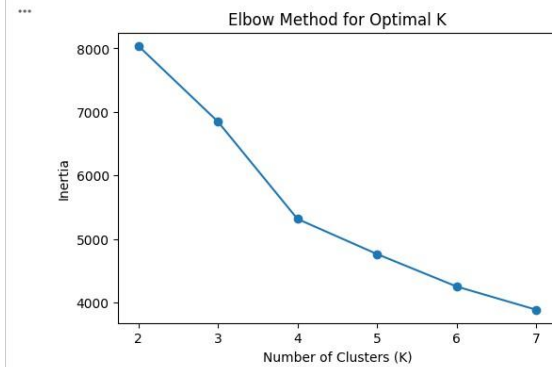


Figure 6: Elbow Method for Optimal K in K-Means Clustering

The K-Means clustering algorithm was applied on the following features: Total_Ratings, Avg_Rating, Gender_Code, and Category_Code. This classifies the fragrance dataset into four clusters of varied demands, which include the following: 1) Very High Demand, 2) High Demand, 3) Trending, and 4) Low Demand.

Cluster Summary:

	Demand_Level	Total_Ratings	Avg_Rating
0	High Demand	2026.911330	4.030350
1	Low Demand	1390.772874	3.829276
2	Trending	1709.890323	4.069734
3	Very High Demand	42024.780488	3.982483

Figure 7: Summary of Demand Levels

The clusters were then labeled manually after the clustering process. The resulting cluster summary is shown in Figure

2.0. The 'Very High Demand' cluster exhibited the highest total ratings of 42,024.78, while the 'High Demand', 'Trending', and 'Low Demand' clusters had total ratings of 2,026.91, 1,709.89, and 1,390.77, respectively. Average customer ratings across clusters ranged from 3.83 to 4.07. The 'Trending' cluster had an average rating of 4.07, but the average rating of 'High Demand' is 4.03, which means fewer customers reviewed this type of fragrance, but those who rated it have more positive feedback.

Recommended Very High Demand New Flavors with Ranking:						
Rank	Flavor	Category	Gender	Avg_Rating	Total_Ratings	
19	1	bergamot	fresh	unisex	4.002202	79761.0
196	2	mandarin orange	fruity	women	3.929352	58415.0
235	3	pear	fruity	women	3.924605	56291.0
404	4	black currant	fruity	women	3.824667	39107.0
130	5	grapefruit	fruity	men	4.049500	31873.0
170	6	lemon	fruity	unisex	4.010235	31223.0
262	7	pink pepper	spicy	unisex	3.968706	30294.0
62	8	cardamom	spicy	unisex	4.018431	28803.0
659	9	saffron	spicy	unisex	4.102754	24201.0
11	10	apple	fruity	men	3.957692	22608.0

Figure 8: Recommended Very High Demand Fragrance Flavors

Figure 8 shows the top 10 fragrances in Very High Demand, ranked by Total_Ratings, which currently are not being offered by Aura Kesidang. These were shortlisted as potential new product recommendations to show which scent categories and gender preferences have a higher probability of succeeding within the market.

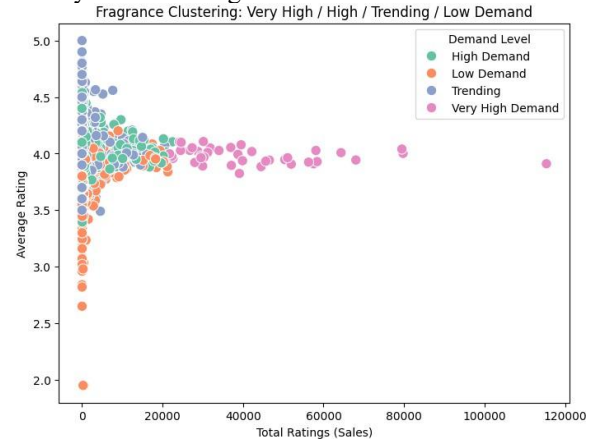


Figure 9: Fragrance Clustering by Demand Level

2) Decision Tree Classification

This is the implementation of a Decision Tree classifier to learn the association of the attributes of fragrances with previously identified levels of demand. The primary goal of this model was to offer an explainable model of the task of forecasting that a new fragrance will be classified as the Very High Demand category.

The categorical attributes (Category and Gender) were utilized as the input of the decision tree in combination with the measures of numbers like Average Rating and Total Ratings. Demand Level is the target variable which was label encoded to convert the categorical levels of demand into numerical numbers.

The shuffling that was done in the dataset was to make sure that the samples were randomly distributed. One-Hot Encoding was used to process the categorical input features. This option is consistent with the best practices, because decision-tree-based models tend to view the category labels as ordinal scales, which may inject an artificial hierarchy. One-Hot Encoding binary indicator columns are used to indicate each unique category, categories are all considered discrete and non-ordered by the model.

```
# One-Hot Encode Category & Gender
ohe = OneHotEncoder(handle_unknown='ignore', sparse_output=False)
x_encoded = ohe.fit_transform(x[['Category', 'Gender']])

# Convert One-Hot back into DataFrame
ohe_cols = ohe.get_feature_names_out(['Category', 'Gender'])
x_encoded = pd.DataFrame(x_encoded, columns=ohe_cols, index=x.index)
x_encoded['Avg_Rating'] = x['Avg_Rating']
x_encoded['Total_Ratings'] = x['Total_Ratings']
```

Figure 10: One Hot Encoding for Categorical Attributes

Once this encoding was done, the resulting One-Hot feature array was concatenated with the numerical variables (Avg_Rating and Total_Ratings), which resulted in the final input dataset to train the model. The data was then subdivided in an 80:20 percentage ratio into the training and test sets. In splitting, stratification was employed to ensure that each category of demand in the sets had the same percentage.

```
# Decision Tree Classifier
dt = DecisionTreeClassifier(max_depth=4, random_state=42)
dt.fit(x_train, y_train)
```

```
DecisionTreeClassifier
DecisionTreeClassifier(max_depth=4, random_state=42)
```

Figure 11: Code snippet for Decision Tree Classification

The Decision Tree Classification was set to use a maximum depth of 4 to tradeoff between the interpretability and the complexity of patterns typical of the various levels of demands. The Decision Tree, therefore offers a hierarchical structure of rules that may be subsequently utilized in the detection of feature combinations that may be indicative of the Very High Demand perfumes.

3) Random Forest Classification

Random Forest classifier was used to further increase predictive stability and also increase the relationship between the features that may be more complex. The same input and output features that appeared in the Decision Tree were used in this model. The Random Forest was set to use 100 estimators with a maximum depth of 6 to ensure that the model is not too complex and it does not overfit.

```
# Fit Random Forest model
rfc = RandomForestClassifier(n_estimators=100, max_depth=6, random_state=42)

rfc.fit(x_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier(max_depth=6, random_state=42)
```

Figure 12: Code snippet for Random Forest Classifier

B. Model Evaluation

The output of the Decision Tree and Random Forest models was assessed with standard performance metrics in the classification, i.e., Accuracy, Precision as well as Recall, calculated separately in training and testing sets.

Accuracy for test set: 0.8984
Accuracy for train set: 0.9038

Precision for test set: 0.9019
Precision for train set: 0.9109

Recall for test set: 0.8984
Recall for train set: 0.9038

Figure 13: Decision Tree model performance evaluation metric

The Decision Tree model had a training accuracy of 0.9038 and test accuracy of 0.8984, with a minimal variation of 0.0054. This small gap shows that the model can be generalize very well. Nevertheless, the two accuracies are less than the accuracy of the Random Forest, and it can be

seen that the Decision Tree is more likely to underfit or overfit compared to the more robust ensemble model.

Precision of the Decision Tree is 0.9109 (train) and 0.9019 (test). Precision is used when the researcher wants to know the percentage of the types of demand correctly identified. As an illustration, the test precision of 0.9019 shows that 90.19% of the labels predicted are accurate. It is a stable performance that has a difference of approximately 0.009, which supports the fact that overfitting is minimal.

Equally, the recall is 0.9038 (train) and 0.8984 (test). Recall is the number of true demands that have been correctly identified by the model (i.e., true positive rate). The recall of 0.8984 shows that the model fits the actual demand categories of the test set with 89.84. Once again, the small gap (≈ 0.005) suggests that the model is consistent but has no ability to describe more complex interactions in the data.

Accuracy for test set: 0.9741
Accuracy for train set: 0.9841

Precision for test set: 0.9744
Precision for train set: 0.9845

Recall for test set: 0.9741
Recall for train set: 0.9841

Figure 14: Random Forest model performance evaluation metrics

In contrast, the Random Forest records much better results. The accuracy of the training is 0.9841, and the test accuracy is 0.9741 with a difference of 0.01. This is a very small gap. The accuracy values are high, indicating the capability to learn and generalize patterns in the data much more. The precision is 0.9845 (train) and 0.9744 (test). Test precision=0.9744 implies that 97.44% of all the predicted classes are correct. This is also an indication that this model has acquired good discriminative boundaries with one-hot encoding.

Recall is 0.9841 (train) and 0.9741 (test), implying that the model is properly able to identify 97.41% of all the actual demand levels. This recall level is an excellent coverage, implying that the model seldom misses the actual class.

In a comparison of the two models, the Random Forest is clearly better than the Decision Tree in all the measures. Though the Decision Tree is stable and slightly overfits, it is much less predictive (accuracy approximately 0.89 -0.90) and underfits compared to the Random Forest.

However, the Random Forest has a score of over 0.97 on accuracy, precision and recall on the test set, that is, it gets more correct, gets fewer misclassified and gets a higher percentage of the true labels.

The reason is that being a combination of multiple decision trees, Random Forest enhances the predictive stability and reduces overfitting compared to using a single Decision Tree, hence, it slightly outperformed on the test set.

Overall, they are able to find the high demand level of fragrances. The decision tree offers explainable features of insights. The predictive accuracy of random Forest is larger since the ensemble prediction model is specifically trained with bagging and random variable selection, reducing the main weakness of the decision trees, overfitting and high variance [1].

Accuracy, Precision, and Recall grouped bar charts of both the training and test sets that provide comparative performance of these models are provided in Figure 15.



Figure 15: Decision Tree vs Random Forest Performance Comparison

C. Visualization and Reporting

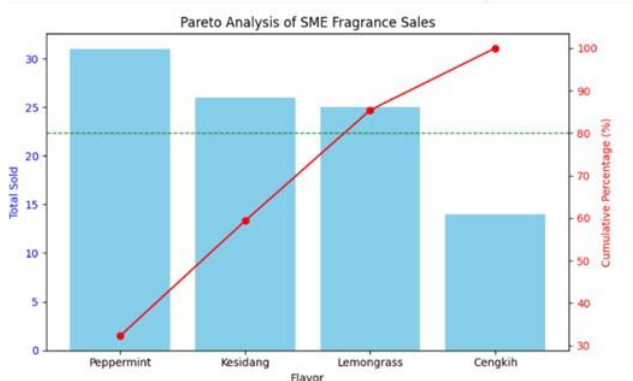


Figure 16: Pareto Chart

For identifying the best-selling fragrance products of SMEs, Pareto analysis was applied to the sales dataset. Sales values were first sorted in a descending manner, which is compatible with the Pareto principle, where a small share of products corresponds to the major contribution. A Pareto chart with a dual axis was created using Matplotlib. The bar plot shows the number of units sold per product, and the complementary cumulative distribution line displays the percentage each fragrance contributes to the total sales volume. Cumulative percentage was calculated by adding each product's sales sequentially and divided by overall sales volume. This clearly indicates that Peppermint (31 units), Kesidang (26 units), and Lemongrass (25 units) together show more than 80% of the cumulative volume, reflecting those variants as core SME products for revenue generation. On the other hand, Cengkih (14 units) has an extremely lower contribution to overall sales, indicating a weak demand from customers. Thus, this Pareto visualization provides clear evidence-based indication on which fragrances need to be emphasized in future production planning, stock allocation, and marketing efforts. Multi-axis Pareto charts could be made by Matplotlib, a grid drawing tool. The library offers a great deal of accuracy in the customization of bar-line combinations, axis scaling, percentage formatting, and reference indicators which allow the visualization to very accurately show the cumulative distribution of product sales. The unproblematic interaction with pandas also makes the efficient handling of preprocessed data possible, which together with Matplotlib's

being an appropriate choice for producing visual outputs that are interpretable and ready for publication results.

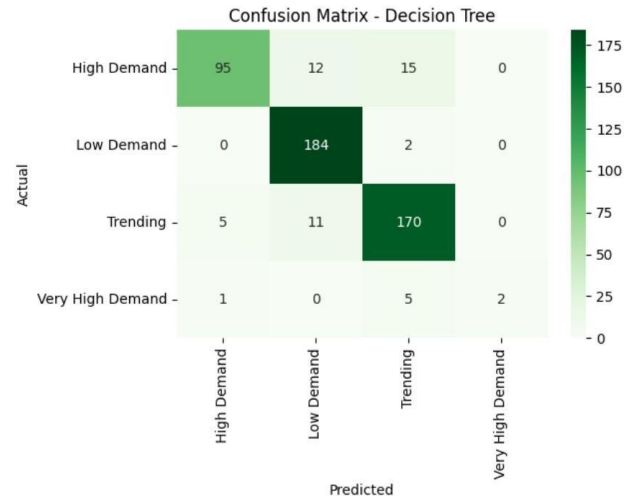


Figure 17: Confusion Matrix: Decision Tree

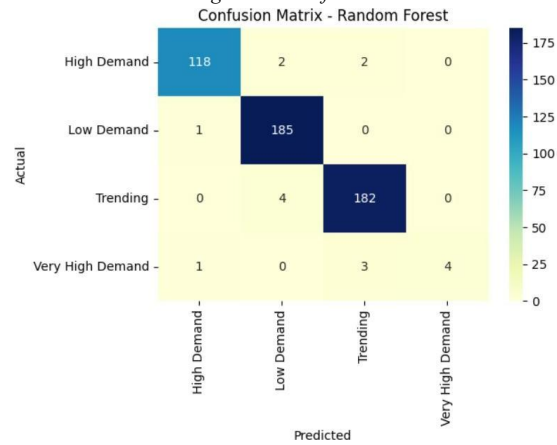


Figure 18: Confusion Matrix: Random Forest

A confusion matrix is used because it shows a class-by-class error visibility, not just overall accuracy.

To generate the confusion matrix, matplotlib was selected because of its flexibility and clean visuals. In addition, a heatmap-style was used to clearly display class-by-class prediction errors, on top of supporting direct visual comparison between the Decision Tree and Random Forest through different colors.

The two figures show the confusion matrices for Decision Tree and Random Forest. The Random Forest demonstrates less confusion, especially within the High and Trending Demand classes, where 118 out of 120 (98.33%) and 182 out of 186 (97.95%) instances respectively were correctly classified. On the other hand, Decision Tree shows more scattered errors. For example, for Trending classification, only 170 out of 192 (88.54%) were correctly classified. This suggests worse generalization for the Decision Tree.

Since a misclassification of high-demand fragrances could lead to missed revenue opportunities, the Random Forest model presents itself to be more reliable for guiding Aura Kesidang's product scaling decisions.

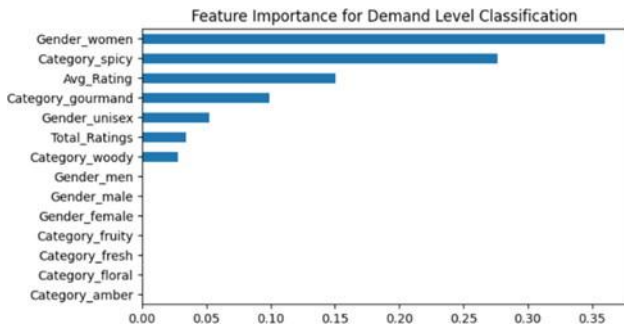


Figure 19: Feature Importance for Demand Level Classification (Decision Tree)

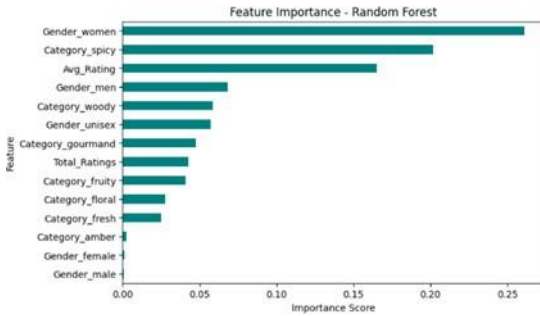


Figure 20: Feature Importance for Demand Level Classification (Random Forest)

To find out which features were the most influential, the prediction of fragrance demand levels in both the classification models was done and during this process the importance of features was analyzed. The Decision Tree and Random Forest both pointed to the same three topmost features: Gender_women, Category_spicy, and Avg_Rating. This indicates that perfumes for women, spicy-scented fragrances, and those with the highest customer ratings are much more likely to be placed in the higher groups of demand. In the Decision Tree model, Gender_women achieved the highest importance value of about 0.36, with Category_spicy and Avg_Rating following. In this case, the model relies greatly on these features for every decision tree node split. The Random Forest model also reflects this pattern, with the Gender_women feature once again being the most important predictor, although it is given a little less weight of about 0.27, because it is the average through many trees.

The Random Forest has an even distribution of importance through additional predictors such as Gender_men, and Gender_unisex, and Category_woody, giving them more importance compared to the Decision Tree model. This shows that this ensemble model is much more capable of uncovering hidden data relationships without depending on a few variables. For both models, the features Category_amber, Gender_female, and Gender_male invariably have meager influence, which means they are not effective in assisting the differentiation of demand levels. Generally, while both models reveal the same top predictors, the Random Forest, because of its ensemble design, renders a much more stable, generalized, and less overfitted representation of feature importance.

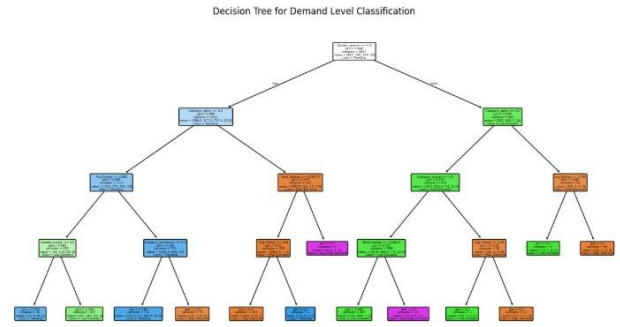


Figure 21: Feature Importance for Demand Level Classification

A decision tree visualization is used because it reveals which feature combinations the model relies on the most when classifying demand levels, instead of merely giving an accuracy value. The simple logic of the decision tree makes it easy to view the main split conditions that drive the prediction on the leaf node. To generate the tree, we utilized Matplotlib alongside scikit-learn's plotting tools, because these allowed us to control layout, fonts, and node-colouring, all of which produces a clear, neat figure that fits this report.

From the decision tree, we observe that the primary split is on the Gender_women feature. This indicates that whether the fragrance is targeted towards women, is a major factor influencing the model's prediction of demand levels. For non-women targeted fragrances, the next key driver is whether the fragrance is spicy. Non-spicy items are mostly classified as Trending or Low Demand, depending on their average rating and whether they are unisex or gourmand – whereas spicy items with very high total ratings (>23357) are classified as Very High Demand, and those below that threshold as High Demand.

For women-targeted fragrances, the tree focuses on spicy vs woody categories, combined with total ratings and average rating. Spicy scents for women with ratings above 3.75 are classified as High Demand, whereas woody scents for women with high ratings (>3.45) or high total rating counts (>24650.5) are classified as High or Very High Demand respectively, while those of low rating fall into Low Demand.

Overall, the tree reveals a set of recurring pattern familiar to the rest of our findings, which is that women-targeted fragrances in spicy or woody scent categories combined with stronger ratings alongside total rating volume are most likely to be of the High Demand class, and in contrast, lower-rated or low-volume variants in less favourable categories tend to be predicted as Trending, or Low Demand.

D. Insights and Recommendations

According to the Pareto analysis, Aura Kesidang currently faces a concentration in demand among their current products (hero fragrances), having Peppermint, Kesidang, and Lemongrass drive majority of the company's demand, while Cengkih contributes very little.

Across clustering, feature importance, and the decision tree, the same pattern occurs – women-targeted + spicy/woody (and selected gourmand categories) + higher ratings / rating volume = higher demand class. However, it's important to note that many of the Very High Demand profiles identified

from the market dataset are not yet in Aura Kesidang's portfolio.

Furthermore, the Decision Tree shows simple, readable rules that match these patterns, while the Random Forest reduces misclassification, especially High and Trending classes, making it far more reliable for being the basis of making real decisions.

Because the model predicts demand levels (Very High / High / Trending / Low), its output can be used directly to guide new fragrance expansions, which would result in marketing and operational decisions made.

Aura Kesidang should leverage the Pareto analysis to focus on producing more stock for the hero fragrances. This is especially crucial to ensure that stock availability does not become a hindrance towards generating potential revenue.

In addition, Aura Kesidang should consider launching 2-3 pilot variants based on the high-potential women-targeted spicy/woody profiles. Weak-demand fragrances (such as Cengkih) should be produced in smaller or niche runs only. This minimizes wasted cost on products that have a higher chance of becoming obsolete.

TikTok/Shopee content should also be utilized to promote high-demand fragrances, using the predicted demand class for a reference to decide how much exposure and budget each one receives.

Operationally, Aura Kesidang should use the Random Forest demand predictions to set pre-packaging and stock levels. For example, the fragrance of class High Demand should set a stock level of x . Moreover, reviews of predictions vs actual sales should be conducted after events to adjust batch sizes.

Finally, Aura Kesidang should keep collecting ratings, sales, and campaign data. These should then be used to retrain the Random Forest model at regular intervals.

V. CONCLUSION

This research paper has been able to successfully develop and validate a data-driven recommendation model that can help Aura Kesidang identify high-potential fragrance segments to expand its business. Combining SME sales data with publicly available fragrance datasets, the analysis not only provided insights through sales metrics but also brought forth dimensions like gender preference, fragrance type, and customer reviews as multi-dimensional indicators. K-Means clustering facilitated the building of a fragrance demand hierarchy of Very High, High, Trending, and Low

which all together provided a systematic way of evaluating the products. The next step of classifying by means of the Decision Tree and Random Forest models revealed that among the major ensemble methods is Random Forest which gives better predictive accuracy, stability, and reliability. With a test accuracy of 97.45%, Random Forest surpasses the Decision Tree (89.44%) and demonstrates better generalization, lower overfitting, and a higher value of precision and recall for all demand categories.

Key insights showed that fragrances targeted for women, from spicy or woody scent categories, and those with higher average ratings are likely to reach high demand. These findings agree with the Pareto analysis that had signaled Peppermint, Kesidang, and Lemongrass as the core revenue drivers, with Cengkih performing dismally in the market.

The project's proposed solution enables the company to set up a scalable, interpretable, and actionable framework for product portfolio optimization, inventory planning, and targeted marketing. Thus, the data-driven approach employed by the SME not only facilitates its slow but sure move from being reactionary to making decisions strategically and based on demand but also finally results in the better allocation of resources that supports the company's sustainable growth.

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REFERENCES

- [1] A. Ali, A. M. Ishaga, H. A. Osman, M. Elhindi, N. Ibrahim, A. Snani, A. Hamid, and A. Hammad, "Machine learning-based residential load demand forecasting: Evaluating ELM, XGBoost, RF, and SVM for enhanced energy system and sustainability," *Science in Information Technology Letters*, vol. 6, no. 1, pp. 1–15, May 2025, doi: 10.31763/sitech.v6i1.1866.